

MAXIMILIAN LIPSKI

Graphics Programmer | Vulkan | C++

+49 15751824418 • lipskidev@outlook.com • maxlipski.dev • linkedin.com/in/maximilian-lipski/ • Dresden, Germany

Summary

Master's student with half a year of experience in machine learning research and real-time rendering, experience in C++ and Vulkan. Key achievements include developing a real-time rendering engine achieving around 400 FPS at 2560x1600 resolution and building an RHI with Vulkan support and future backend expansions.

Experience

TU Dresden - ScaDS.AI — <http://scads.ai>

Dresden, DE

Student Researcher

11/2025 - 04/2026

- Researched machine learning approaches to accelerate geometric rendering of geometric objects
- Developed and trained neural networks for geometric approximation problems using CUDA-enabled machine learning workflows
- Investigated neural representations of Hermite spline tubes as signed distance functions
- Trained a neural network for ray-intersection with Hermite spline tubes
- Designed and evaluated experiments to analyze the accuracy and performance of learned geometric approximations
- Achieved promising results for fixed primitives while identifying parameterization as a major challenge or generalization

Education

Technical University Dresden

Dresden, DE

Master of Science, Computer Science

10/2024 - 01/2027

- Grade: 1.6
- Thesis: Rendering of transparent lines using hardware-accelerated linear swept sphere primitives
- Active, Expected Graduation: 01/2027

Heinrich-Heine University

Düsseldorf, DE

Bachelor of Science, Computer Science

10/2020 - 09/2024

- Grade: 1.9
- Thesis: Development of a Skew-Heap Data Structure for Persistent Memory

Skills

Programming: C/C++ • GLSL • CUDA • SPIR-V • Multithreading

Graphics APIs: Vulkan • OpenGL

Tools: RenderDoc • NVIDIA Nsight Graphics • NVIDIA Nsight Systems • Git • CMake • Premake • Unreal Engine 5

Platforms: Linux • Windows

Research: Machine learning • Neural networks

Projects

Rodan - Real-Time Rendering Engine

- Created a real-time rendering engine built on top of a custom Rendering-Hardware Interface (RHI) with a Vulkan backend
- Achieves around 400 FPS while rendering the Sponza scene at 2560x1600 on an RTX 5070 with one directional light and tone mapping
- Built a physically based rendering (PBR) pipeline with image-based lighting (IBL)
- Developed an HDR tone mapping pipeline
- Added glTF 2.0 asset loading with support for KHR_materials_transmission and KHR_materials_volume
- Integrated Dear ImGui for in-engine tooling and debugging
- Debugged and profiled rendering workloads using NVIDIA Nsight Graphics and RenderDoc

Velos - RHI

- Solely designed and implemented a backend-independent RHI with a Vulkan backend
- Developed abstractions for pipelines, descriptor sets, textures, samplers, buffers, and command lists
- Built automatic SPIR-V shader reflection for descriptor set and pipeline layout generation
- Architected the system to support future graphics backends such as DirectX 12
- Implemented automatic SPIR-V reflection with automatic pipeline creation

Professional Development

 Game Industry Conference (GIC) Sprint
2026 – Katowice, Poland

Attended talks on game engine technology,
rendering, and graphics programming.

Languages

German Native

Polish Native

English C1